

EXDA Metadata Report

Project: VH2D-Project

Generated: 2026-05-27 22:43:19

Includes: Plan, DAQ Systems, Sensors Mapping, Gas Mixing

Responsible Researcher: PhD Student Javier I. Camacho

Project Objective:

LU / DBI / Kistler pressure DAQ systems characterisation.

Project Description:

The primary objectives of the experiments are:

1. Internal Over-pressure Development

Measurement of internal over-pressure development during vented hydrogen deflagrations in a partially confined enclosure for homogeneous flammable hydrogen-air mixtures with hydrogen concentrations up to 19 vol.%.

2. Flame Propagation (Under Consideration)

Measurement of flame propagation characteristics, including potential estimation of flame time-of-arrival (TOA) at selected locations.

3. Instrumentation Evaluation

Evaluation and cross-comparison of different pressure measurement systems and data acquisition configurations, including sensor characteristics and the influence of measurement chain dynamics (e.g., time constant (τ) effects, sampling rate).

4. Measurement Uncertainty Factors

Investigation of experimental factors that may affect the accuracy and reliability of pressure measurements, including structural vibration effects and thermal protection methods.

5. Venting Behaviour

Investigation of vent opening behaviour and the influence of vent rupture mechanisms on pressure development, including comparison between pressure-driven thin-film rupture and controlled vent opening using a hot-wire cutting system.



EXDA Plan Export - Experiment_Plan_v000

Responsible Researcher: PhD Student Javier I. Camacho

Objective: LU / DBI / Kistler pressure DAQ systems characterisation.

Start: 2026-05-25 | Deadline: 2026-05-29

Total Experiments: 24

Generated: 2026-05-27 22:43:18

Group: Prep

Group Objective: Preparation

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
Prep-01	Yes	2026-05-25	13.5	back	Thin Plas.	101325	292.15	3	Prep-01/DAQ-1/Prep-01.exp/data/Prep_01-DAQ-1.tpc5 Prep-01/DAQ-2/Prep-01-DAQ-2.mf4 Prep-01/H2CM/Prep-01-H2CM-U.csv

Group: Verif

Group Objective: Performance Verification

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
Verif-01	Yes	2026-05-26	13.5			101325	293	5	Verif-01/DAQ-1-Verif-01.tpc5 Verif-01/DAQ-2-Verif_01/DAQ-2-Verif_01.mf4 Verif-01/H2CM-L-260526-120036.csv Verif-01/H2CM-M-260526-115955.csv Verif-01/H2CM-U-260526-115817.csv
Verif-02	Yes	2026-05-26	14			101325	293	3	Verif-02/DAQ-1-Verif-02.tpc5 Verif-02/DAQ-2-Verif_02/DAQ-2-Verif_02.mf4 Verif-02/H2CM-U-M-L-260526-144155.csv
Verif-03	Yes	2026-05-26	16			101325	293	3	Verif-03/DAQ-1/DAQ-1-Verif-03.tpc5 Verif-03/DAQ-2/DAQ-2-Verif-03/DAQ-2-Verif-03.mf4 Verif-03/H2CM/H2CM-U-M-L-Verif-03-260526-160316.csv
Verif-04	Yes	2026-05-26	18			101325	293	3	Verif-04/DAQ-1/DAQ-1-Verif-04.tpc5 Verif-04/DAQ-2/DAQ-2-Verif-04/DAQ-2-Verif-04.mf4 Verif-04/H2CM/H2CM-U-M-L-Verif-04-260526-174304.csv

Group: VH2D-01

Group Objective: Vented deflagration concentration series (Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
-----	------	----------	--------	----------	------	---------	--------	-------	-----------------------------

VH2D-01-01	Yes	2026-05-27	16	Back	Thin Plas.	101325	293	3	VH2D-01-01/DAQ-1/DAQ-1-VH2D-01-01.tpc5 VH2D-01-01/H2CM/H2CM-U-M-L-VH2D-01-01-260527-140320.csv VH2D-01-01/DAQ-2-3/DAQ-2-3-VH2D-01-01.mf4
VH2D-01-02	Yes	2026-05-27	16	Back	Thin Plas.	101325	293	3	VH2D-01-02/DAQ-1/DAQ-1-VH2D-01-02.tpc5 VH2D-01-02/H2CM/H2CM-U-M-L-VH2D-01-02-260527-150043.csv VH2D-01-02/DAQ-2-3/DAQ-2-3-VH2D-01-02.mf4
VH2D-01-03	Yes	2026-05-27	18	back	Film	101325	293	2	VH2D-01-03/DAQ-1/DAQ-1-VH2D-01-03.tpc5 VH2D-01-03/DAQ-2-3/DAQ-2-3-VH2D-01-03.mf4
VH2D-01-04	Yes	2026-05-27	18	back	thin film	101325	293	3	VH2D-01-04/DAQ-1/DAQ-1-VH2D-01-04.tpc5 VH2D-01-04/DAQ-2-3/DAQ-2-3-VH2D-01-04.mf4 VH2D-01-04/H2CM/H2CM-U-M-L-VH2D-01-04-260527-164423.csv
VH2D-01-05	No	2026-05-27	19	back	Thin Plas.	101325	293	0	-
VH2D-01-06	No	2026-05-27	19			101325	293	0	-
VH2D-01-07	No	2026-05-27	20			101325	293	0	-
VH2D-01-08	No	2026-05-27	20			101325	293	0	-

Group: VH2D-02

Group Objective: Vented deflagration concentration series (NO Hotwire)

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-02-01	No	2026-05-28	12			101325	293	0	-
VH2D-02-02	No	2026-05-28	14			101325	293	0	-
VH2D-02-03	No	2026-05-28	16			101325	293	0	-
VH2D-02-04	No	2026-05-28	18			101325	293	0	-

Group: VH2D-03

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-03-01	No	2026-05-28	12			101325	293	0	-
VH2D-03-02	No	2026-05-28	14			101325	293	0	-
VH2D-03-03	No	2026-05-28	16			101325	293	0	-
VH2D-03-04	No	2026-05-28	18			101325	293	0	-

Group: VH2D-04

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-04-01	No	2026-05-29	18			101325	293	0	-

Group: VH2D-05

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-05-01	No	2026-05-29	18			101325	293	0	-

Group: VH2D-06

Group Objective: -

Run	Done	Schedule	H2 (%)	Ignition	Vent	P0 (Pa)	T0 (K)	Files	File Path (Raw_Data Folder)
VH2D-06-01	No	2026-05-29	18			101325	293	0	-

EXDA DAQ Systems Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total DAQ Systems: 5

Reference PDF: LU-DBI Measurement Chain (frontend/public/LU-DBI_MeasurementChain_v001.pdf)

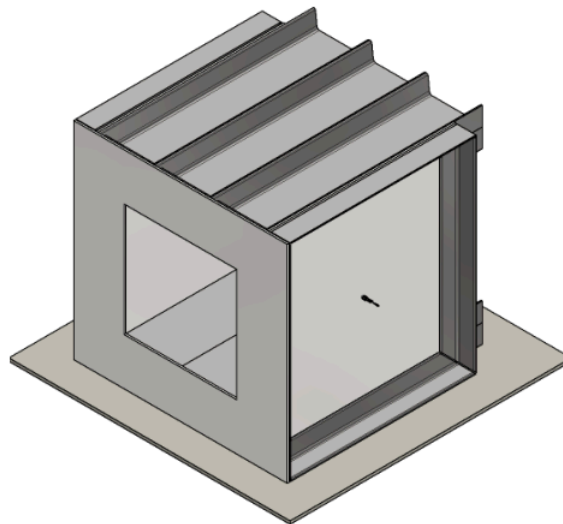
Reference PDF: Mixture Sampling Sub-System (frontend/public/MixtureSampling_Sub-System_000.pdf)

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Name	Quantity	Vendor	Model	Serial	Fs (Hz)	Ch	Owner	Last Cal.	Cal Cert ID	Active	Notes
DAQ-1	pressure	Elsys -	TraNET	441309	-	8	LU	2024-10-21	-	Yes	2 equipment
		Kistler	204S8								
DAQ-2	pressure	Kistler	5165A4	6306100	-	4	DBI	-	-	Yes	-
DAQ-3	pressure	Kistler	5165A4	5358310	-	4	Kistler	-	-	Yes	-
DAQ-4	pressure	Kistler	-	-	-	8	Kistler	-	-	Yes	-
H2CM	concentration	JCT	JTCD-300	14362	1	-	LU	2025-07-21	-	Yes	-

DAQ Reference Diagram - LU-DBI Measurement Chain

Source: frontend/public/LU-DBI_MeasurementChain_v001.pdf

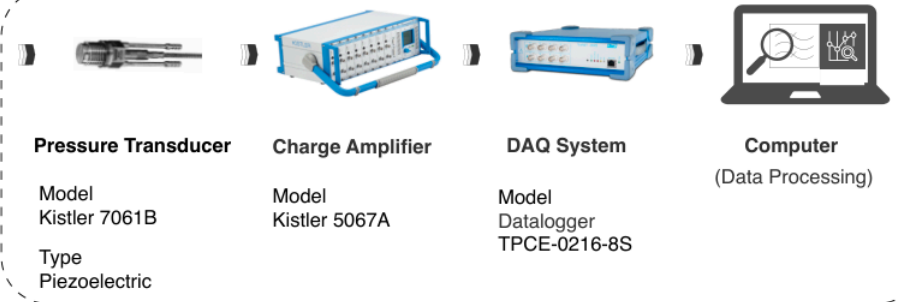


Explosion Chamber

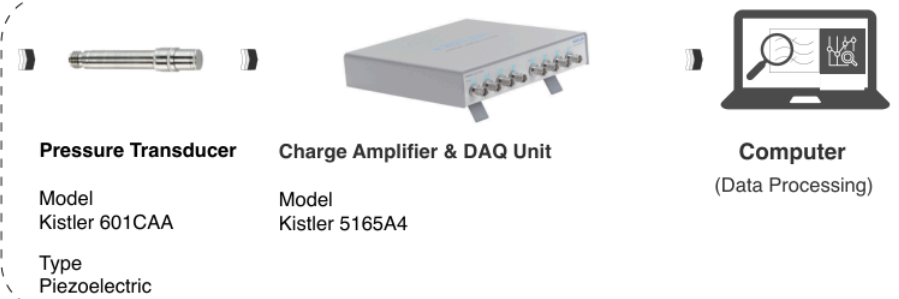
Dimensions (Internal): 900 x 900 x 900 [mm]

Material: Steel

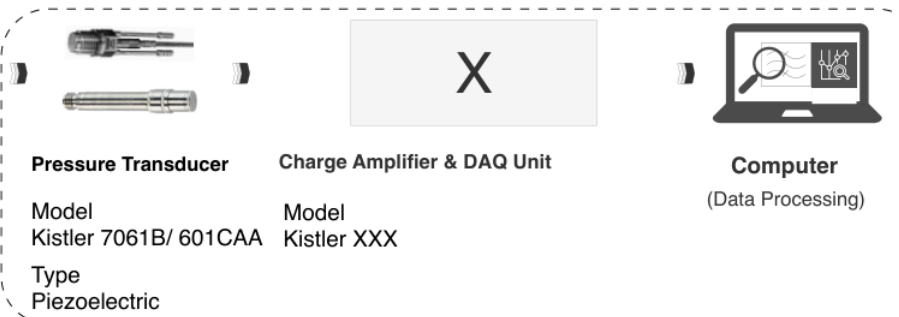
LU-Measurement Chain



DBI-Measurement Chain

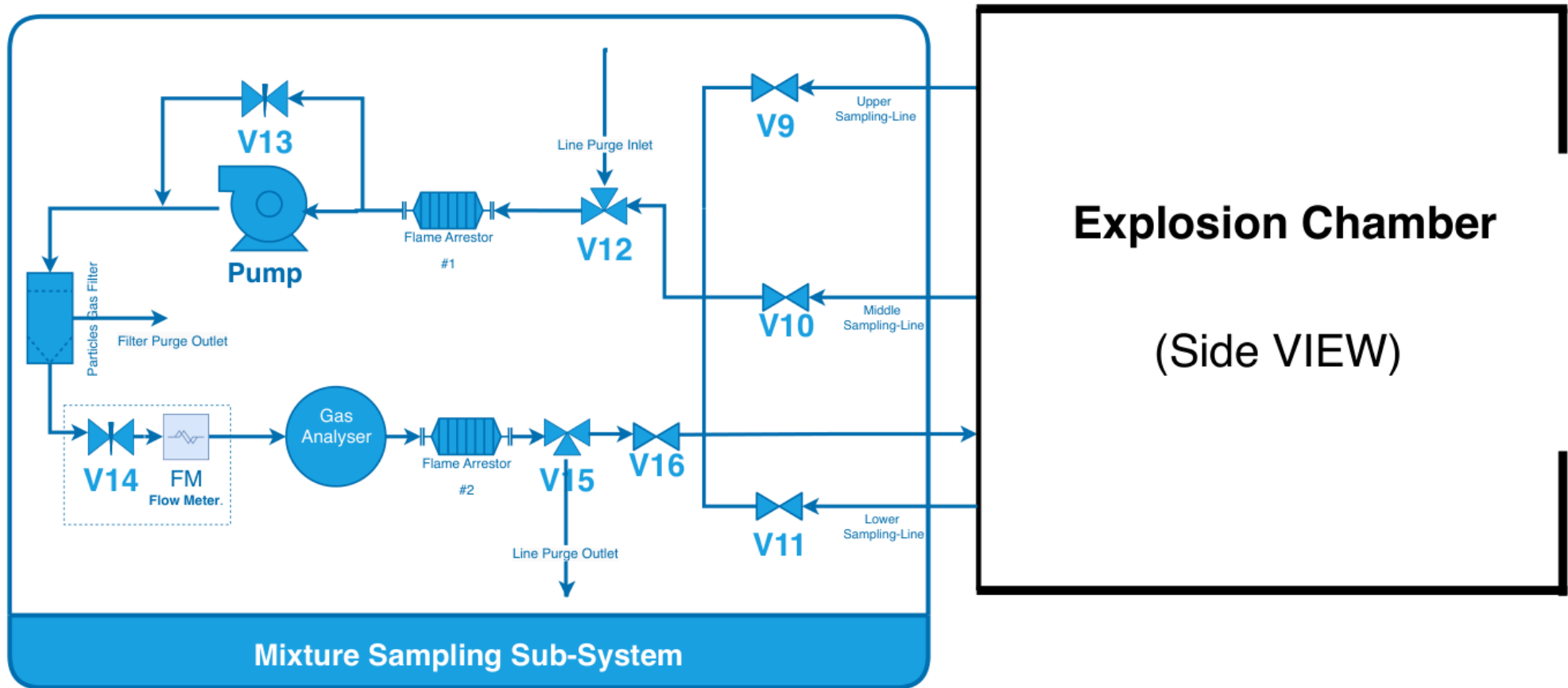


Kistler-Measurement Chain



DAQ Reference Diagram - Mixture Sampling Sub-System

Source: frontend/public/MixtureSampling_Sub-System_000.pdf



EXDA Sensors Mapping Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total Groups: 8

Total Sensor IDs: 18

Total Sensor Mappings: 33

Sensor Mounting Reference File: frontend/public/SensorMountingLocation-000.pdf

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Group: Prep

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivity	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
Prep	P1	pressure	DAQ-1	CH1	1254024	-80.88 pC/bar	5L	(-0.45,0.399,0.593)	flush	Yes	No	-	Complete
Prep	P2	pressure	DAQ-2	CH2	6266801	-37.02 pC/bar	6L	(-0.450,0.403,0.290)	flush	Yes	No	-	Complete
Prep	Trig-1	voltage	DAQ-1	CH5	-	- pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete
Prep	Trig-2	voltage	DAQ-2	CH4	-	- pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete

Group: Verif

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivity	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
Verif	P1	pressure	DAQ-1	CH1	1254024	-80.88 pC/bar	5L swaped to 3L on Verif-04	(0.45,0.593,0.396)	flush	No	No	-	Complete
Verif	P2	pressure	DAQ-2	CH1	6266801	-37.02 pC/bar	6L	(0.450,0.290,0.403)	flush	Yes	No	-	Complete
Verif	P3	pressure	DAQ-1	CH2	1008977	-81.58 pC/bar	3L swaped to 5L on Verif-04	(-0.45,0.396,0.427)	other	Yes	Yes	-	Complete
Verif	P4	pressure	DAQ-2	CH2	6266802	-37.39 pC/bar	4L	(-0.45,0.416,0.447)	flush	Yes	No	-	Complete
Verif	P5	pressure	DAQ-1	CH3	1946319	-79.12 pC/bar	3L	(-0.45,0.416,0.447)	flush	Yes	No	-	Complete
Verif	P6	pressure	DAQ-2	CH3	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Incomplete
Verif	P7	pressure	DAQ-2	CH3	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Incomplete

Verif	P8	pressure	DAQ-3	CH1	-	- pC/bar	-	(-, -, -)	flush	Yes	No	-	Incomplete
Verif	PP0	pressure	Kistler	CH6	-	- pC/bar	-	(0, 2, 0.68)	flush	Yes	No	-	Incomplete
			LabAmp										
Verif	PP15	pressure	Kistler	CH7	-	- pC/bar	-	(0, 2, 0.68)	flush	No	No	-	Incomplete
			LabAmp										
Verif	PP30	pressure	Kistler	CH8	-	- pC/bar	-	(0, 2, 0.68)	flush	No	No	-	Incomplete
			LabAmp										
Verif	Trig-1	voltage	DAQ-1	CH5	-	- pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete
Verif	Trig-2	voltage	DAQ-2	CH4	-	- pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete
Verif	Trig-3	voltage	DAQ-3	CH2	-	- pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete

Group Reference Note: Verification

=====
Group: VH2D-01

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi-	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
VH2D-01	P1	pressure	DAQ-1	CH1	1254024	-80.88 pC/bar	5L swaped to 3L on Verif-04	(0.45,0.593,0. 396)	flush	No	No	-	Complete
VH2D-01	P2	pressure	DAQ-2	CH1	6266801	-37.02 pC/bar	6L	(0.450,0.290,0 .403)	flush	Yes	No	-	Complete
VH2D-01	P3	pressure	DAQ-1	CH2	1008977	-81.58 pC/bar	3L swaped to 5L on Verif-04	(-0.45,0.396,0 .427)	flush	Yes	No	-	Complete
VH2D-01	P4	pressure	DAQ-2	CH2	6266802	-37.39 pC/bar	4L	(-0.45,0.416,0 .447)	recessed	Yes	Yes	-	Complete
VH2D-01	P5	pressure	DAQ-1	CH3	1946319	-79.12 pC/bar	3L	(-0.45,0.416,0 .447)	recessed	Yes	Yes	-	Complete
VH2D-01	P6	pressure	DAQ-2	CH3	6266803	-37.66 pC/bar	2L	(-0.450,0.376, 0.447)	recessed	Yes	No	-	Complete
VH2D-01	P7	pressure	DAQ-2	CH4	931538	-80.73 pC/bar	1R	(0.45,0.381,0. 447)	recessed	Yes	No	-	Complete
VH2D-01	P8	pressure	DAQ-3	CH1	6266804	-37.55 pC/bar	4R	(0.450,0.361,0 .433)	recessed	Yes	No	-	Complete
VH2D-01	P9	pressure	DAQ-1	CH1	1008976	-78.79 pC/bar	1L	(-0.45,0.396,0 .467)	recessed	Yes	No	-	Complete
VH2D-01	PP1	pressure	Kistler	CH6	-	- pC/bar	-	(0, 2, 0.68)	other	Yes	No	-	Incomplete
			LabAmp										

VH2D-01	PP2	pressure	Kistler	CH7	-	-	pC/bar	-	(0,2,0.68)	other	No	No	-	Incomplete
			LabAmp											
VH2D-01	PP3	pressure	Kistler	CH8	-	-	pC/bar	-	(0,2,0.68)	other	No	No	-	Incomplete
			LabAmp											
VH2D-01	Trig-1	voltage	DAQ-1	CH5	-	-	pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete
VH2D-01	Trig-2	voltage	DAQ-2	CH4	-	-	pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete
VH2D-01	Trig-3	voltage	DAQ-3	CH2	-	-	pC/bar	-	(-, -, -)	N/A	Yes	No	M-Duino	Complete

Group Reference Note: Copied from Verif: Verification

=====
Group: VH2D-02

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
VH2D-02	-	-	-	-	-	-	-	-	-	-	-	-	Reference

=====
Group: VH2D-03

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
VH2D-03	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Group Reference Note: same configuration as preparation group

=====
Group: VH2D-04

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
VH2D-04	-	-	-	-	-	-	-	-	-	-	-	-	Reference

=====
Group: VH2D-05

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
VH2D-05	-	-	-	-	-	-	-	-	-	-	-	-	Reference

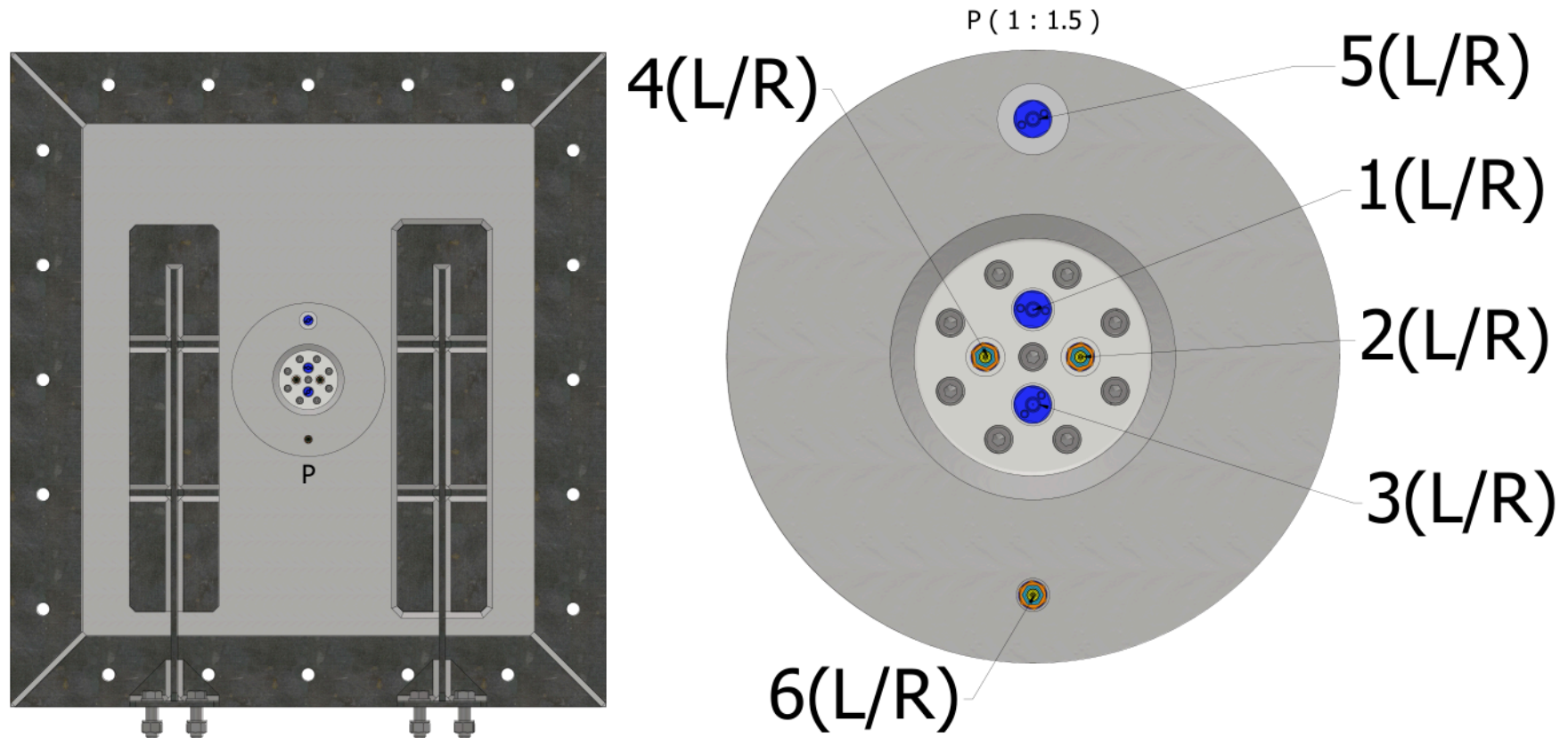
=====
Group: VH2D-06

Group	Sensor	Qty	DAQ	Ch	S/N	Sensitivi	Location	Coord.(x,y,z)m	Mount	Active	Blind	Trig Method	Status
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
VH2D-06	-	-	-	-	-	-	-	-	-	-	-	-	Reference

Sensor Mounting Reference Diagram

Source: frontend/public/SensorMountingLocation-000.pdf

Pressure Sensor Location Mapping Overview



EXDA Gas Mixing Export

Project: VH2D-Project

Responsible Researcher: PhD Student Javier I. Camacho

Total Records: 11

Generated: 2026-05-27 22:43:19

MATLAB Verified: Yes

Verification File A: scripts/GasMixingVerificationFiles/H2_MFC_Fill_Calculator_v000.m

Verification File B: scripts/GasMixingVerificationFiles/AuxFcn_H2_MFC_FillCalculator_000.m

Shared Chamber Setup (applies to all runs unless noted otherwise):

L=0.900 m, W=0.900 m, H=0.900 m, Pipes+=VARIES m³, Hotwire+=0.000 m³, Welded-=0.000 m³, Bolts-=0.000 m³, Vcorr=VARIES L

Note: setup values vary across runs; detailed per-run setup table is included below.

Core Gas Mixing Results

Group	Run/Test	H2%	Pch(Pa)	Tch(K)	H2(g)	H2inj(L)	MFC	t(s)	t(min)	Updated
Prep	Prep-01	13.50	101325	292.15	8.276	98.41	20.00	301.3	5.0	2026-05-25 17:27:49
Verif	Verif-01	13.50	101325	294.25	8.217	98.42	20.00	299.2	5.0	2026-05-26 11:46:08
Verif	Verif-02	14.00	101325	293.15	9.384	111.97	20.00	334.8	5.6	2026-05-26 14:03:57
Verif	Verif-03	16.00	101325	294.15	10.638	127.37	20.00	387.3	6.5	2026-05-26 15:53:24
Verif	Verif-04	18.00	101325	291.65	12.070	143.29	20.00	439.5	7.3	2026-05-26 17:31:49
VH2D-01	VH2D-01-01	16.00	101325	293.15	10.674	127.37	20.00	388.6	6.5	2026-05-27 13:51:19
VH2D-01	VH2D-01-02	16.00	101325	292.05	10.714	127.37	20.00	390.1	6.5	2026-05-27 14:47:32
VH2D-01	VH2D-01-03	18.00	101325	291.35	12.083	143.29	20.00	439.9	7.3	2026-05-27 15:24:55
VH2D-01	VH2D-01-04	18.00	101325	290.85	12.103	143.29	20.00	440.7	7.3	2026-05-27 16:30:34
VH2D-01	VH2D-01-05	19.00	101325	290.55	12.789	151.26	20.00	465.6	7.8	2026-05-27 17:50:04
VH2D-01	VH2D-01-07	20.00	101325	290.56	13.462	159.22	20.00	490.1	8.2	2026-05-27 17:57:37

Per-Run Chamber Setup (because values vary)

Group	Run/Test	L (m)	W (m)	H (m)	Pipes+ (m3)	Hotwire+ (m3)	Welded- (m3)	Bolts- (m3)	Vcorr(L)	Notes
Prep	Prep-01	0.900	0.900	0.900	0.000	0.000	0.000	0.000	729.00	-
Verif	Verif-01	0.900	0.900	0.900	0.000	0.000	0.000	0.000	729.00	-
Verif	Verif-02	0.900	0.900	0.900	0.071	0.000	0.000	0.000	799.80	-
Verif	Verif-03	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
Verif	Verif-04	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-01	VH2D-01-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-01	VH2D-01-02	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-01	VH2D-01-03	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-01	VH2D-01-04	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-

VH2D-01		VH2D-01-05		0.900		0.900		0.900		0.067		0.000		0.000		0.000		796.08		-
VH2D-01		VH2D-01-07		0.900		0.900		0.900		0.067		0.000		0.000		0.000		796.08		-

